Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/08/2022
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neuronal and Behavioral Response to Exposure to HZE Particles		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	08/01/2016	**End Date:**	07/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2022 per L. Stephenson/Space Radiation (Ed., 8/2/21) NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:	May 2020 report: CoInvestigator (CoI) Shibu Poulouse, Ph.D., is no longer CoI on the project.		
COI Name (Institution):	Shukitt-Hale, Barbara Ph.D. (US Department of Agriculture/Agricultural Research Service, Beltsville, MD)		
Grant/Contract No.:	NNX16AE06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to single, above threshold doses of HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neuronal and behavioral functioning after the conclusion of the mission. The effects of exposure to multiple, subthreshold doses of the same or different particles or protons have not been extensively studied. The objectives of the proposed research program are: (1) to evaluate the effects of repeated below threshold/above threshold exposures to the same and different HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) experiments currently underway will examine the capacity of dietary antioxidants (blueberries) administered after exposure to 56Fe particles to ameliorate the disruptive effects of irradiation on cognitive performance and changes in neurochemical functioning. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our data indicate that there is a significant risk that exposure to low, NASA-relevant doses of HZE particles produces alterations in CNS function and in the behaviors that are dependent upon the integrity of specific CNS systems. However, current research suggests several caveats that might modulate the effectiveness of such exposures in disrupting neuronal function and cognitive performance. The experiments presented in this report were designed to evaluate several factors which may affect the degree to which exposure to space radiation might affect the cognitive ability of astronauts on exploratory class missions. The first series of experiments utilized exposure to 4He and 56Fe particles in male and female rats. The second series of experiments utilized exposure to 4He and 1H particles because these particles will constitute the major source of radiation within the spacecraft during exploratory class missions. The first series of experiments examined the effects of repeated exposure to a below threshold dose of 4He particles on a subsequent exposure to either a below or above threshold dose 56Fe particles separated by 72 hr on cognitive performance in male and female rats. The experimental design attempts to mimic the most likely pattern of radiation exposure to which astronauts will be exposed on exploratory class missions and involves repeated exposures to below threshold doses. In this context, threshold is defined as the lowest dose of a specific particle which will produce a deficit in cognitive performance. The results indicated that the effects of combined exposures varied as a function of both the specific task (operant responding, novel object or novel spatial performance) and the sex of the subject. A second series of experiments evaluated the effects of exposure to 4He followed by exposure to 1H or exposure to 1H followed by exposure to 4He particles separated by 2 hrs. Because the majority of the dose experienced by astronauts within the spacecraft would be provided by these particles, the order of irradiation was alternated; some rats received 4He particles 2 hr prior to exposure to 1H and some received 1H prior to 4He particles. The effects of repeated exposures on cognitive performance were task dependent and also dependent upon the order of presentation. A final series examined the capacity of dietary antioxidants (blueberries) administered before and/or after exposure to 56Fe particles to ameliorate the disruptive effects of irradiation on cognitive performance and changes in neurochemical functioning. The results indicated that placing subjects on a diet containing 2% blueberry extract for 45 days prior to exposure to 56Fe particles or after exposure were equally effective in preventing the radiation-induced alteration of neuronal function (oxidative stress and neuroinflammation) and the associated disruption of cognitive performance. These results support the use of dietary antioxidants as effective countermeasures for the effects of exposure to space radiation.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Rabin BM, Miller MG, Shukitt-Hale B. "Effects of preexposure to a subthreshold dose of helium particles on the changes in performance produced by exposure to helium particles." Life Sci Space Res (Amst). 2023 May;37:88-96. https://doi.org/10.1016/j.lssr.2023.03.003 ; PMID: 37087183 , May-2023
Articles in Peer-reviewed Journals	Cahoon DS, Rabin BM, Fisher DR, Shukitt-Hale B. "Effects of HZE-particle exposure location and energy on brain inflammation and oxidative stress in rats." Radiat Res. 2023 Sep 27. https://doi.org/10.1667/RADE-22-00041.1 ; PMID: 37758038 , Sep-2023
Articles in Peer-reviewed Journals	Cahoon DS, Fisher DR, Rabin BM, Lamon-Fava S, Wu D, Zheng T, Shukitt-Hale B. "Galactic cosmic ray particle exposure does not increase protein levels of inflammation or oxidative stress markers in rat microglial cells in vitro." Int J Mol Sci. 2024 May 29;25(11):5923. https://doi.org/10.3390/ijms25115923 ; PMID: 38892109; PMCID: PMC11172496 , May-2024